

Olist E-Commerce Sales & Customer Analytics Project Documentation

1. Project Overview

The Olist E-Commerce Project focuses on analyzing customer purchasing behavior, sales performance, customer segmentation, and revenue trends using SQL and Power BI.

The project aims to transform raw e-commerce data into meaningful business insights through data cleaning, SQL analysis, KPI tracking, and interactive dashboards.

2. Business Objectives

The major objectives of this project are:

- Analyze overall sales and customer performance
 - Identify top-performing customers and states
 - Track monthly revenue trends
 - Perform RFM (Recency, Frequency, Monetary) customer segmentation
 - Understand customer purchasing patterns
 - Build interactive dashboards for business decision-making
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3. Dataset Information

Dataset Source:

Brazilian E-Commerce Public Dataset by Olist

The dataset contains information related to:

- Customers
- Orders
- Payments
- Products
- Reviews
- Order Items

Main tables used in the project:

- customers_dataset
 - orders_dataset
 - order_payments_dataset
 - order_items_dataset
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4. Tools & Technologies Used

Tool	Purpose
MySQL	Data cleaning and analysis
Power BI	Dashboard creation and visualization
Excel/CSV	Dataset storage
SQL Window Functions	Customer ranking and analysis

5. SQL Analysis Performed

The following SQL operations were performed:

Data Cleaning

- Checked NULL values
- Removed duplicates
- Verified data consistency

Business Analysis

- Monthly revenue trend analysis
- Revenue by state
- Top customers analysis
- Customer ranking using RANK()
- RFM segmentation analysis

Advanced SQL Concepts Used

- GROUP BY
 - JOINS
 - Aggregate Functions
 - CASE WHEN
 - Window Functions
 - ORDER BY
 - Common filtering techniques
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6. KPI Metrics Created

The following KPIs were created in Power BI:

Sales Dashboard KPIs

- Total Customers
- Total Orders
- Total Revenue

RFM Dashboard KPIs

- Average Recency
 - Average Frequency
 - Total Revenue
 - Customer Count
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7. Dashboard Pages

Page 1: Sales & Customer Analytics Dashboard

This dashboard includes:

- Monthly Revenue Trend
- Revenue by State
- Top Customers
- Customer Table
- State Filter
- Business Insights

Key Insights

- SP state generated the highest revenue
 - Revenue peaked during May–August
 - Sales are concentrated in major states
 - Top customers contribute significantly to total sales
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Page 2: RFM Customer Dashboard

This dashboard includes:

- Customer Segment Distribution
- Revenue by Customer Segment
- Top Customers by Revenue
- Customer Frequency vs Revenue Scatter Plot
- Customer Segment Filter
- RFM Insights

Key Insights

- Lost Customers segment contains the highest number of customers
 - Loyal and Champion customers contribute significantly to revenue
 - A small customer group generates a major share of revenue
 - Higher frequency does not always result in higher revenue
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8. RFM Segmentation Explanation

RFM stands for:

- Recency → How recently a customer purchased
- Frequency → How often a customer purchases
- Monetary → How much revenue the customer generates

Customers were segmented into:

- Champions
 - Loyal Customers
 - Potential Loyalists
 - Regular Customers
 - At Risk Customers
 - Lost Customers
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9. Project Outcome

This project helped in:

- Understanding customer purchasing behavior
 - Identifying high-value customers
 - Monitoring business performance
 - Improving analytical and dashboarding skills
 - Applying SQL and Power BI concepts on real-world data
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10. Conclusion

The Olist E-Commerce Analytics Project demonstrates practical skills in SQL, data analysis, customer segmentation, and dashboard development.

The project provides valuable business insights and supports data-driven decision-making using interactive visualizations and analytical techniques.